

Learning Bayesian Game Families, with Application to Mechanism Design

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1. Game-family learning enables generalization across the parameter space, supporting more effective mechanism optimization.
2. Learned interim models extrapolate more accurately and approximate equilibria more reliably than ex ante.
3. Learned interim models support the generation of beneficial piecewise best-response strategies.

Link to Paper:



Motivation & Methodology

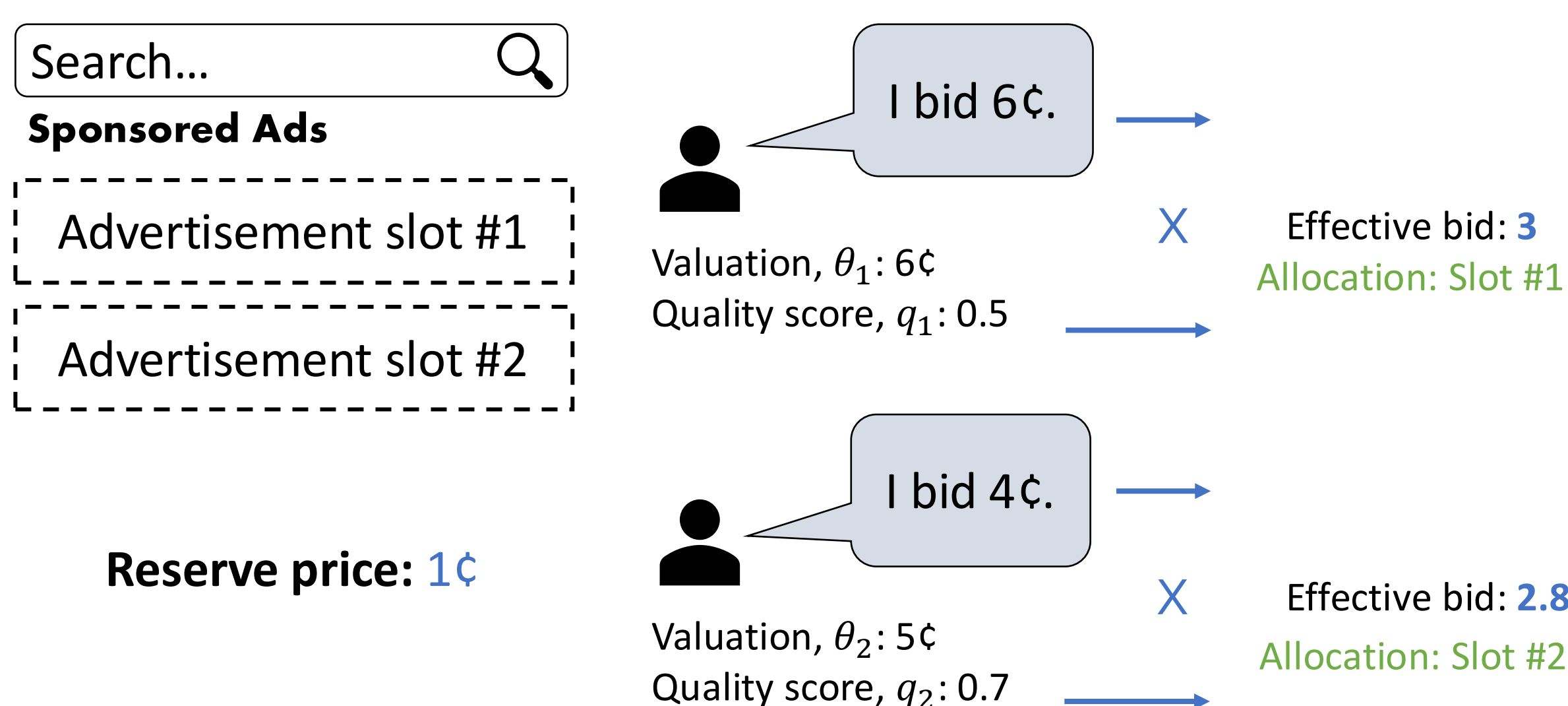
Empirical Mechanism Design

Goal: Find the parameter setting that optimizes the design objective in equilibrium.

Knowledge Source: Agent-based simulation model

Challenge: Limited resources for simulation queries

Example: Online Advertising Auctions



Players have **private information** that informs their choice of action and influences their payoff.

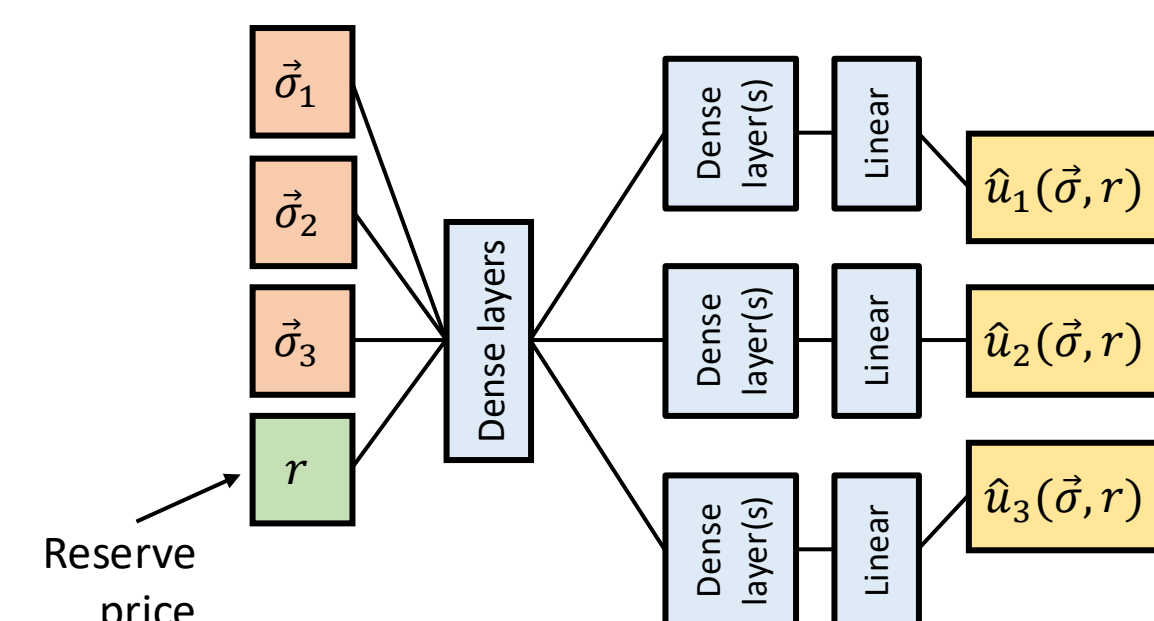
Note: In our experiments, the auction is dynamic: advertisers may attempt to revise bids in a second stage, with attempts succeeding probabilistically.

Methodology Overview

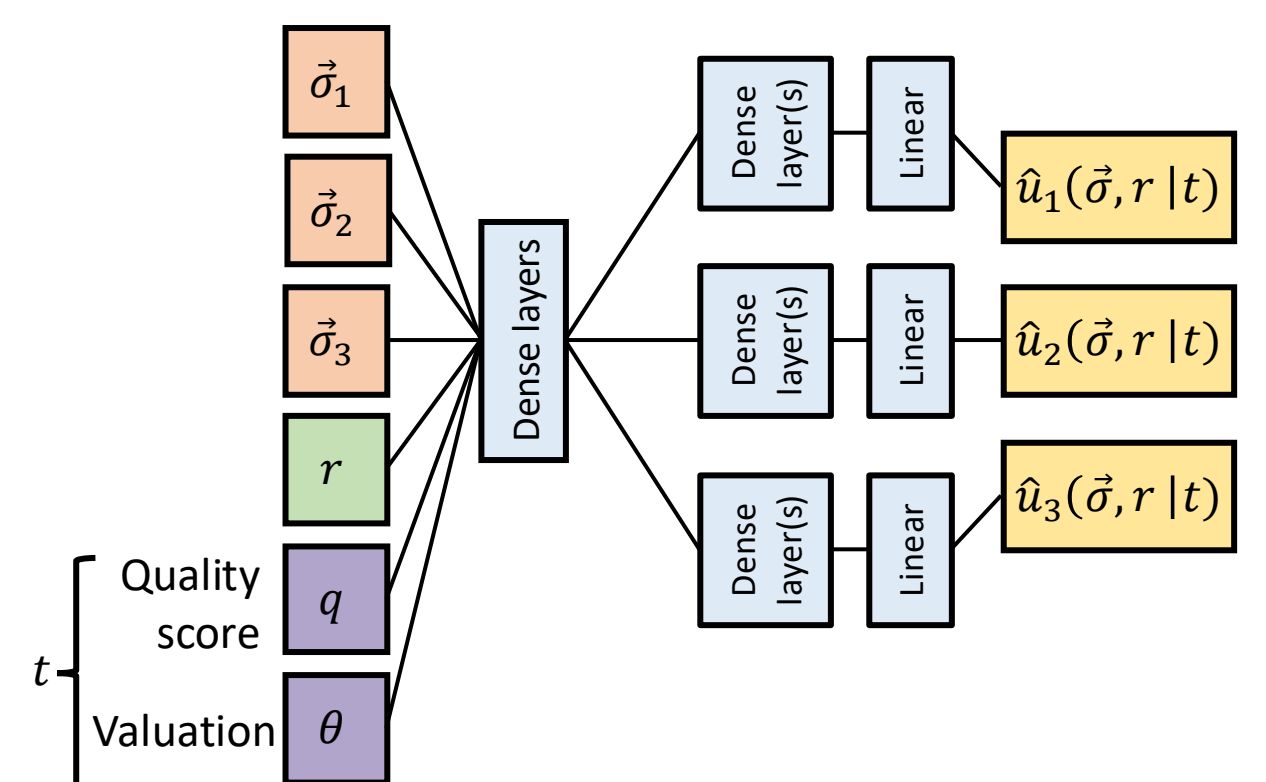
Extend game-family learning [Gatchel and Wiedenbeck, 2023] to the Bayesian setting, exploiting the type-conditional form of strategies.

Learning & Analyzing Bayesian Game Families

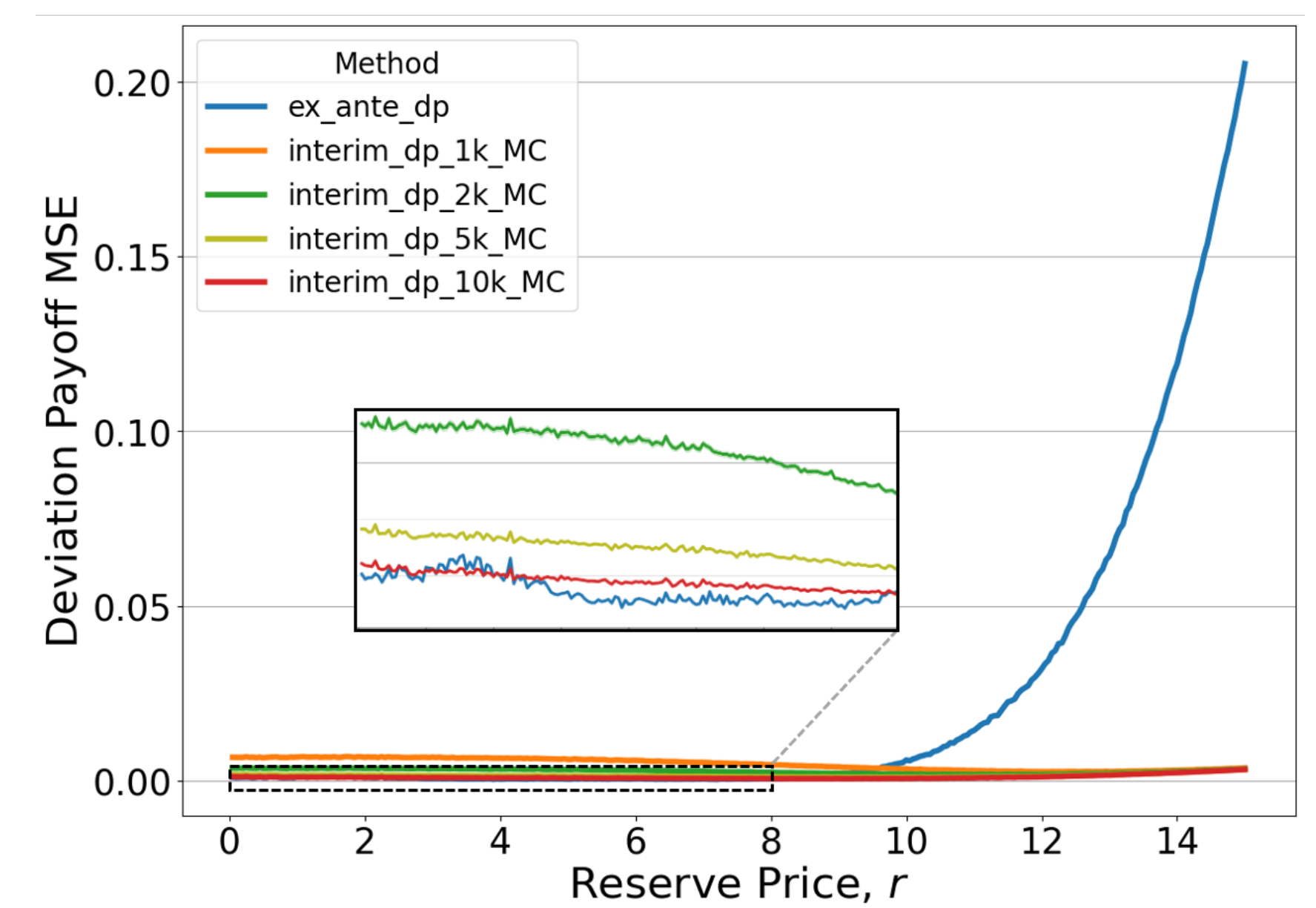
Ex Ante Deviation Payoffs



Interim Deviation Payoffs



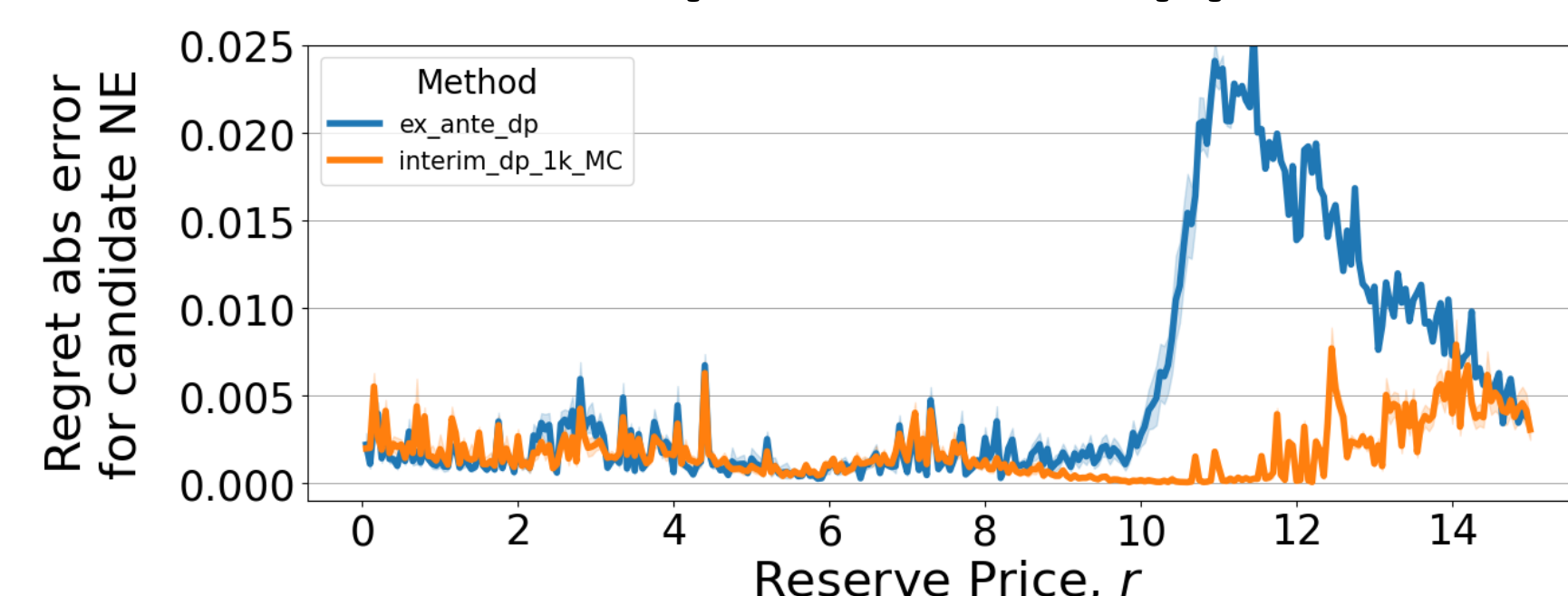
Model Learning Results



With enough marginalization samples, interim model accuracy matches ex ante.

$r > 8$: Interim but not ex ante models extrapolate well beyond the trained range.

Equilibrium Approximation Results



The interim method can better distinguish candidate equilibria from non-convergent mixtures.

Empirical Mechanism Design (EMD) with Learned Game Families

Grid Optimization

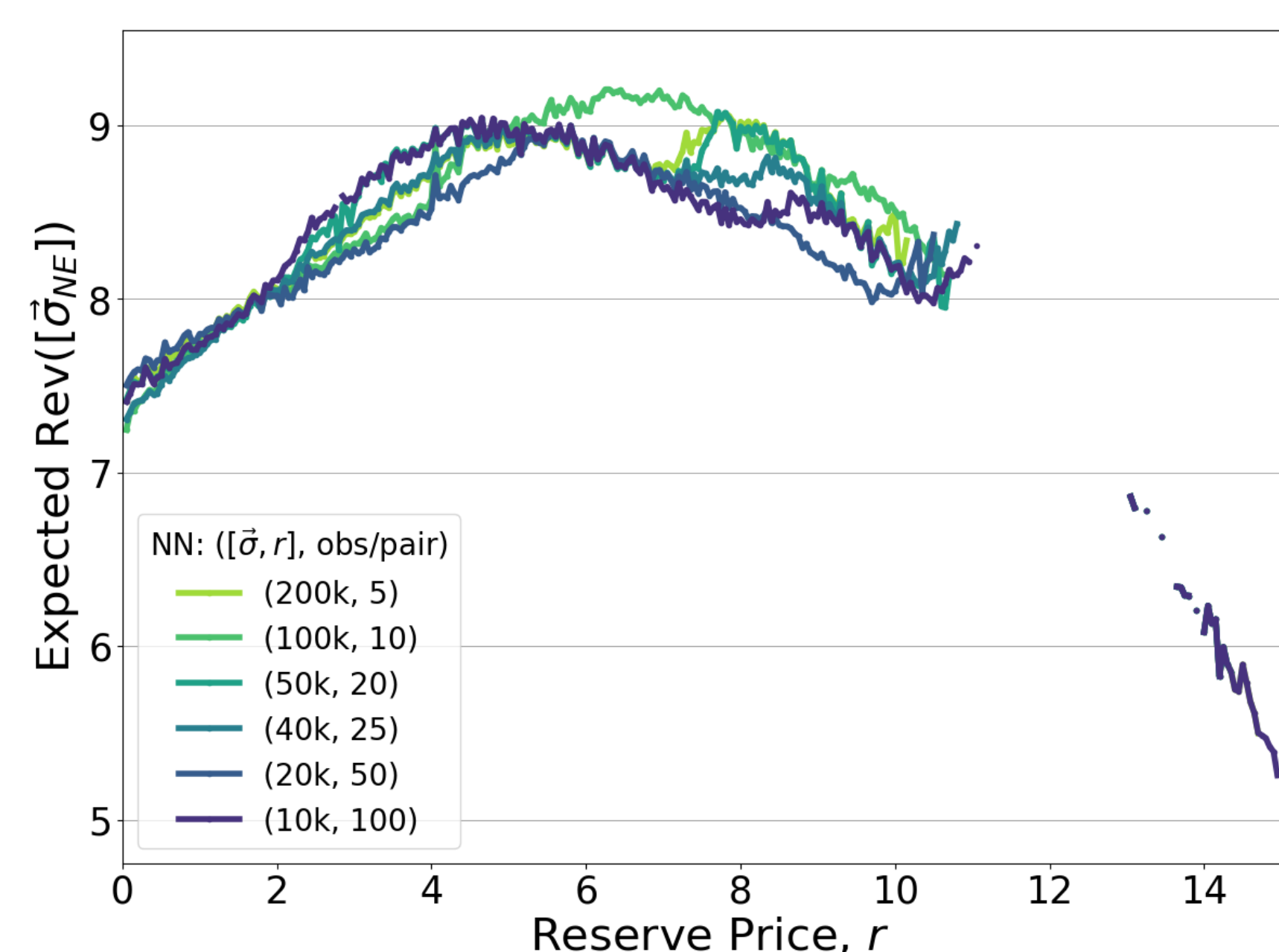
For each $\Gamma(r)$ and learned model:

- Use learned model to identify equilibria
- Compute expected revenue, averaged across all confirmed equilibria

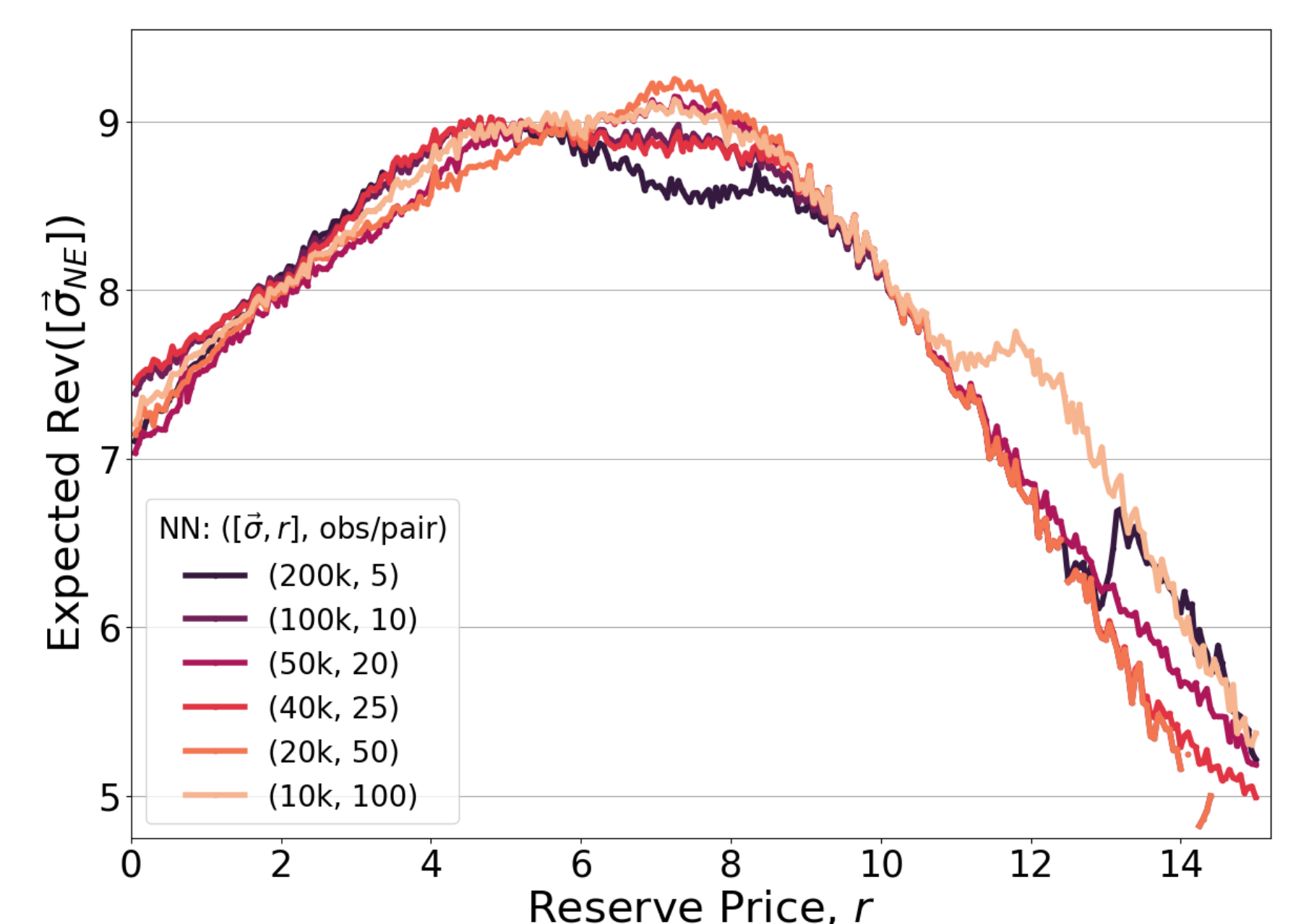
Curves from (L) ex ante models exhibit more discontinuities than (R) interim models.

(Both) Expected revenue curves from model to model may vary significantly.

Ex Ante



Interim



Type-Conditional Piecewise Best-Response Strategies

Piecewise Best-Response (BR) Strategies

Motivation:

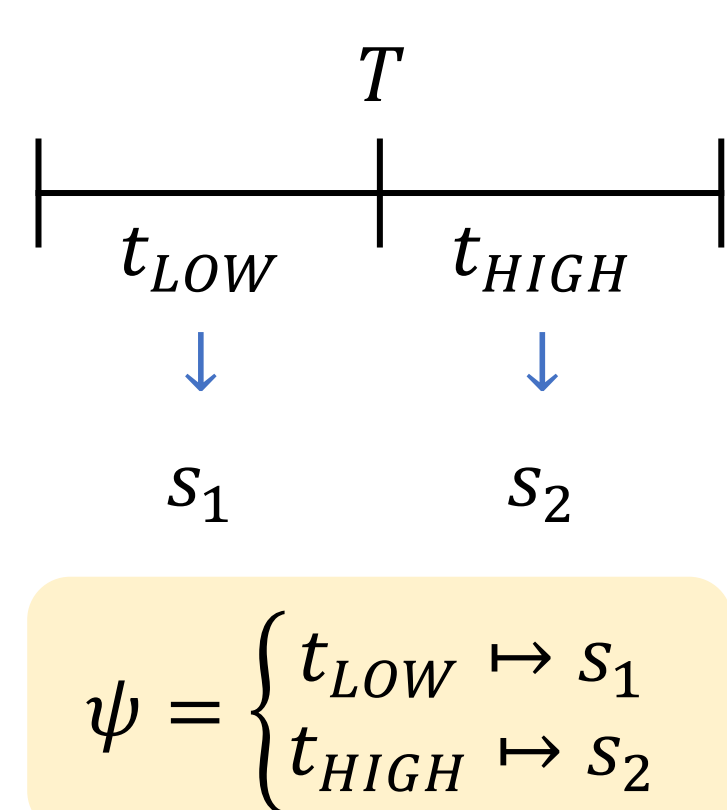
S is a highly restricted subset of all possible strategies. What if an omitted strategy is a beneficial deviation?

Example:

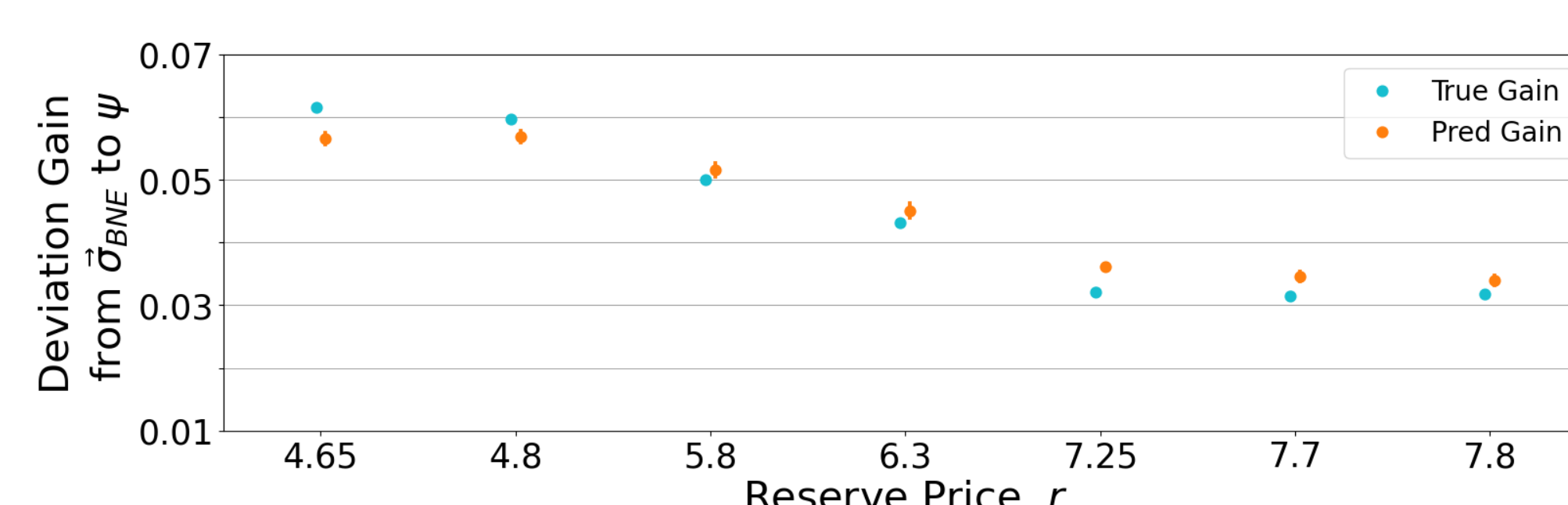
- $T = \{t_{LOW}, t_{HIGH}\}$

S	
s_1	Bid truthfully
s_2	Shade bid

- Ex ante BNE: s_1

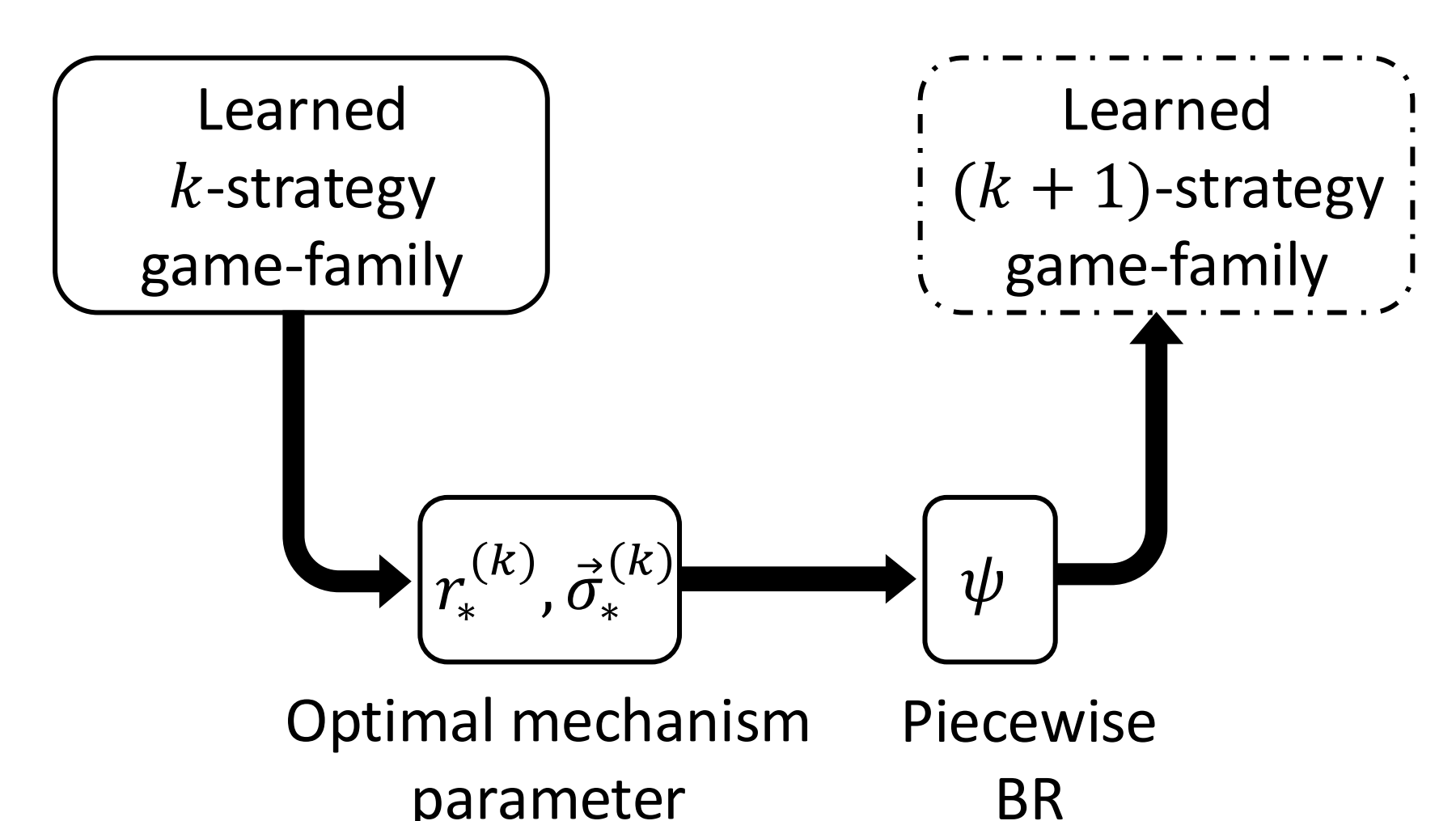


Payoff Gain by Playing Piecewise BRs



Profitable deviation refutes equilibrium over baseline strategies at candidate mechanism optimum.

Future Work: Iterative EMD



...repeat until no beneficial piecewise deviation emerges.